

Lab 3: The Moon's Diameter and its Distance to Earth

Before you start, beware that most of the following illustrations are deceiving! Actually, you can't represent in a readable way at the same scale the Earth, the Moon and the Sun, and at the same time the distances Earth-Sun or Earth-Moon:

For instance, if we represent the Earth by a 1-cm diameter disk, then the Moon's diameter would be 3 mm and it would be 30 cm away from the Earth. But the Sun would then be 120 *m* far from the Earth, and its diameter would be more than 1*m* !

0.1. The Principle of the Lunar Eclipse

A lunar eclipse occurs when the Moon passes through the Earth's shadow. Since the

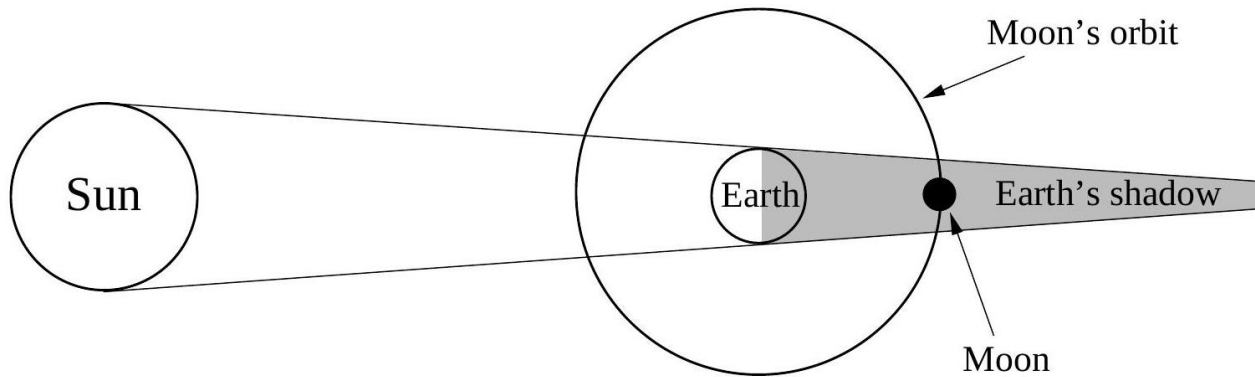


Figure 1: The principle of the Lunar Eclipse.

orbital plane of the Moon is tilted 5° out of the ecliptic (the orbital plane of the Earth) eclipses do not occur at every Full Moon. Most of the time the Moon is passing either "above", or "below" the shadow of the Earth. But if conditions are right (the Moon is very nearly on the ecliptic at full Moon) you can see one of nature's most spectacular phenomena a lunar eclipse.

Interactive exercises: To understand the necessary conditions for a lunar eclipse please use the interactive tools under the title "The cause of a Lunar Eclipse" in the web page of this exercise. To see the progress of an actual lunar eclipse play the movie "Evolution of a Total Lunar Eclipse" in the web page of this exercise.

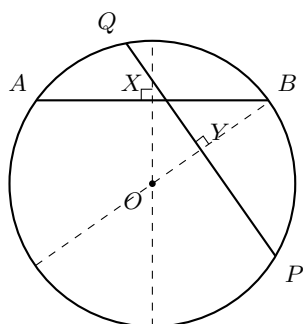
The goal of this exercise is to compute the diameter of the Moon and its distance from Earth using photographs taken during a lunar eclipse. At the end of this laboratory exercise you will find pictures of the Lunar Eclipse which occurred on October 27, 2004. You will use these pictures for your calculations.

0.2. The Diameter of Earth's Shadow

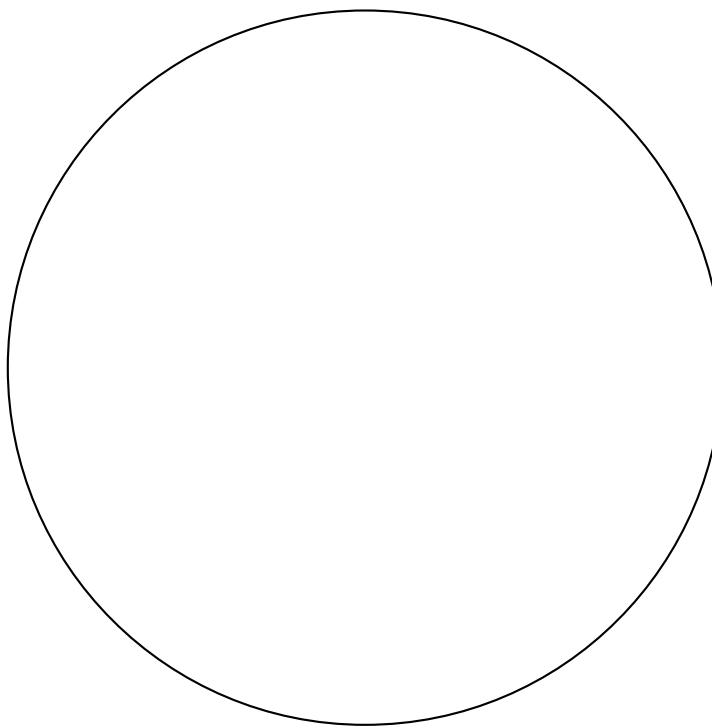
In the first step you will determine how much bigger the diameter of Earth's shadow (the shadow of the Earth which is projected on the surface of the Moon) is than the Moon's diameter. For simplicity consider the shadow of the Earth to be circular. Also, denote the diameter of Earth's shadow by S and the diameter of the Moon by M . In order to determine $\frac{S}{M}$ follow the steps outlined below.

Prelab Requirement: Chords and Perpendicular Bisectors

To find the middle point of a circle, we could just guess, however for big distances like the distance to the Moon, a simple guess could end up being thousands of miles off. Instead we will use a special property of circles to find the center. This is following in the footsteps of the ancient Greek mathematicians who first studied geometry. The property is that the perpendicular bisector of **any chord**, no matter where you choose to draw it, passes through the center of the circle. By drawing two different chords and their perpendicular bisectors, we can find the center of the circle as the intersection of the two bisectors. A **chord** is a line segment whose endpoints both lie on a circle. A **perpendicular bisector** of a segment is a line that crosses the segment at its midpoint and makes a 90° angle with the segment.



Example



Student circle

Prelab requirement: two chord perpendicular bisectors meet at the center of the circle.

Before lab, complete the blank circle above.

1. Use a ruler to draw two different chords on the blank circle. Do not make either chord go through the middle of the circle.
2. Measure the length of each chord with a ruler and mark its midpoint.
3. Place the center mark of your protractor on the midpoint of the chord. Line up the 0° – 180° baseline of the protractor with the chord.
4. Mark the 90° direction, then draw a straight line through the midpoint and your 90° mark. This is the perpendicular bisector.
5. Repeat for the second chord. The two perpendicular bisectors should meet at the center of the circle; label that point O .

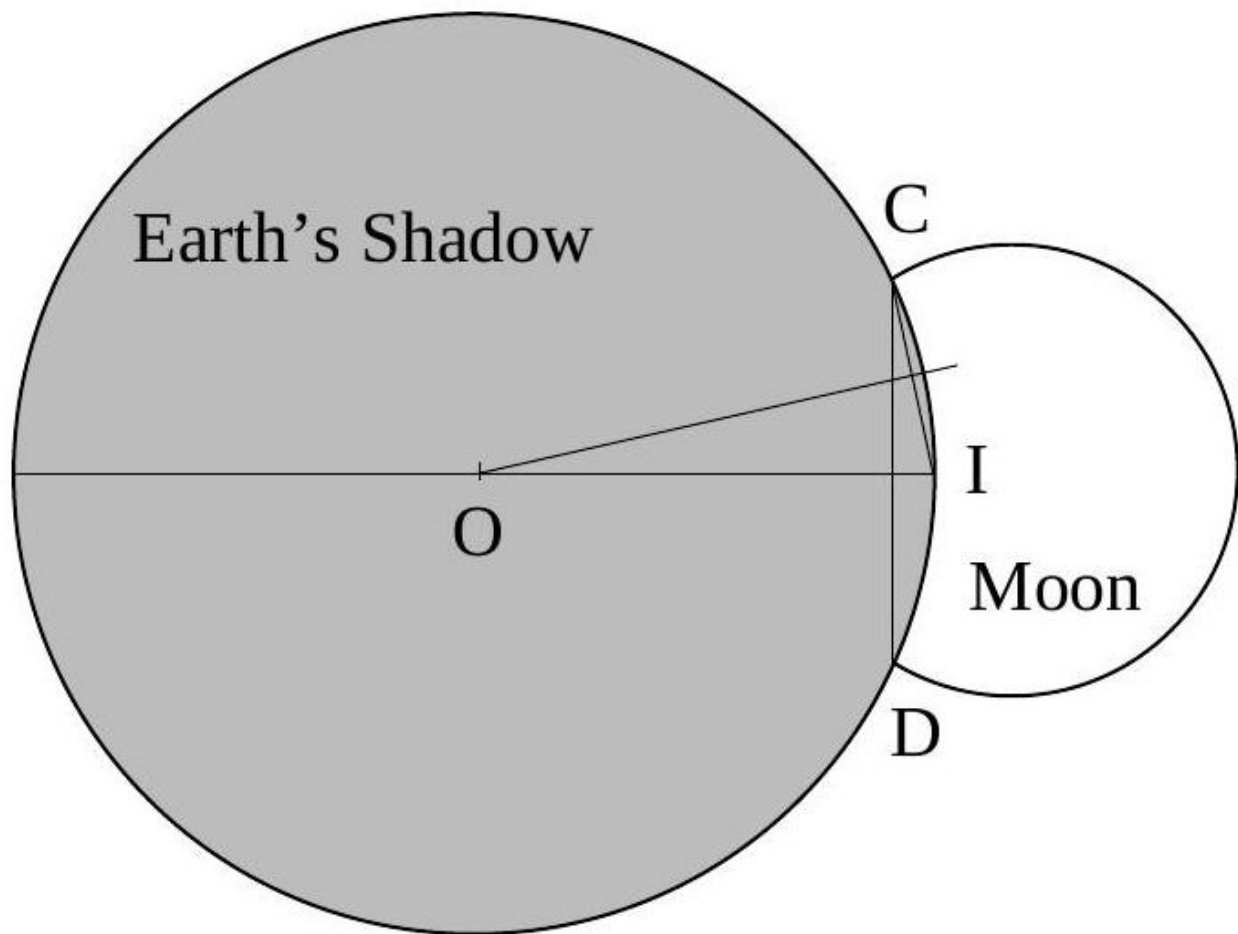


Figure 2: Geometric construction used to determine the diameter of Earth's shadow. This is the view from Earth of the Moon during a lunar eclipse, as Earth's shadow passes over the Moon.

1. Connect points **C** and **D**, these are the points where the shadow of the Earth intercepts the circumference of the Moon. Draw the perpendicular bisector of chord **CD**; the bisector will cross the shadow of the Earth at point **I**. Connect points **C** and **I**. Draw the perpendicular bisector of chord **CI**; this will cross the perpendicular bisector of chord **CD** at the center of the shadow of Earth, point **O**. (See Figure 2.)
2. After determining the center of Earth's shadow measure its radius and from that calculate the diameter S ; record your value next to the picture in Figure 6. Also, measure the diameter of the Moon M from the picture and record it there. Now, calculate the ratio of S and M for each picture in Figure 6 and record your value next to the corresponding picture.
3. Finally, calculate the average ratio of S and M from your measurements in Figure 6 and record the value here.
The average ratio, $\frac{S}{M} = \underline{\hspace{2cm}}$ A typical value of this ratio is about 2.2, which means that the shadow of the Earth at the distance of the Moon is about 2.2 times bigger than the Moon itself.

Diameter of the Moon

Now, we will use our previous result ($\frac{S}{M} = 2.2$) to calculate the true diameter of the Moon. In the calculation we will use that the diameter of the Earth is $E = 12,700$ km.

We consider the shadow of Earth as a cone (see Figure 1), and we compute its diameter at the distance of the Moon. There are no big mathematical difficulties here, but some of you may find this method a little difficult to understand. If it is so, please do not hesitate to ask us.

It can be measured that the apparent diameter of the Sun is almost the same as the apparent diameter of the Moon (about half a degree). To convince yourself you need to observe a solar eclipse and see the disk of the Moon covering exactly the disk of the Sun.

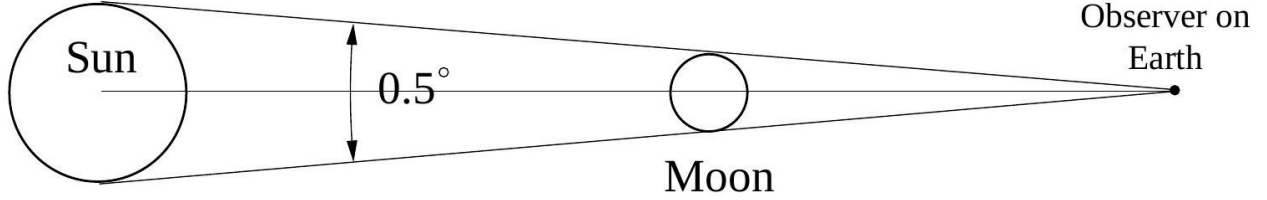


Figure 3: Apparent diameter of the Moon and the Sun as seen from Earth.

Using this information we can set up the diagram shown in Figure 4 for a lunar eclipse. On the diagram please note the line connecting the "lower" edge of the Sun and the "lower" edge of the Earth, thereafter referred to as **CD**, the line connecting the "upper" edge of the Sun and the "upper" edge of the Earth, thereafter referred to as **AB**, and the line connecting point **P** on the Sun and the "lower" edge of the Earth, thereafter referred to as **PD**.

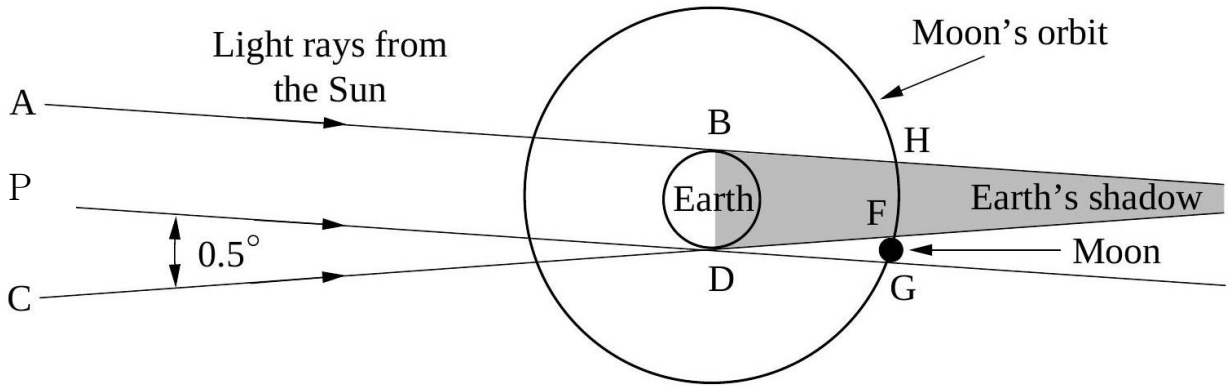


Figure 4: Diagram of a Lunar Eclipse as seen from above Earth.

We can consider **AB** and **PD** to be parallel, since the distance to the Sun is so big (the Earth-Sun distance is 400 times as big as the Earth-Moon distance and 12,000 times as big as the diameter of Earth). The angle $\angle FDG$ is equal to $\angle PDC$, i.e. 0.5° , which is the same thing as the diameter of the Moon seen at the distance of the Moon. Thus we can conclude that the distance **FG** is equal to the diameter of the Moon M .

Using the argument that **AB** and **PD** are parallel, from Figure 4 we can read out the following relations

$$\mathbf{HG} = \mathbf{BD} = E = \text{Earth's diameter} ,$$

and

$$\mathbf{HG} = \mathbf{HF} + \mathbf{FG},$$

which using the diameters can be written as

$$E = S + M.$$

Now using our result from the previous section, $\frac{S}{M} = 2.2$, we can calculate the diameter of the Moon. Since $\frac{S}{M} = 2.2$ we can write

$$S = 2.2M.$$

Substituting this into our previous equation

$$S + M = E = 12,700 \text{ km}$$

we obtain that

$$2.2M + M = 3.2M = 12,700 \text{ km}$$

After solving this equation for M we get that $M = 3,969 \text{ km}$ and $S = 8,731 \text{ km}$. In fact, the precise value of the diameter of the Moon $M = 3,476 \text{ km}$ is very close to our calculated value.

How close is your value? Use the ratio you obtained for $\frac{S}{M}$ in the previous section and repeat the calculation above.

Calculate your value for M ; record it here and in Tables I and II.

$M = _____ \text{ km}$

The Distance to the Moon

Our last step is to calculate the distance from the Earth to the Moon using our previous results. As you can see in Figure 5, the diameter of the Moon is approximately equal to that part of the Moon's orbit which is seen under 0.5° angle from Earth. We also know that the complete orbit of the Moon is seen under 360° angle from Earth. Using these facts please compute the circumference of the Moon's orbit; Table I will help you.

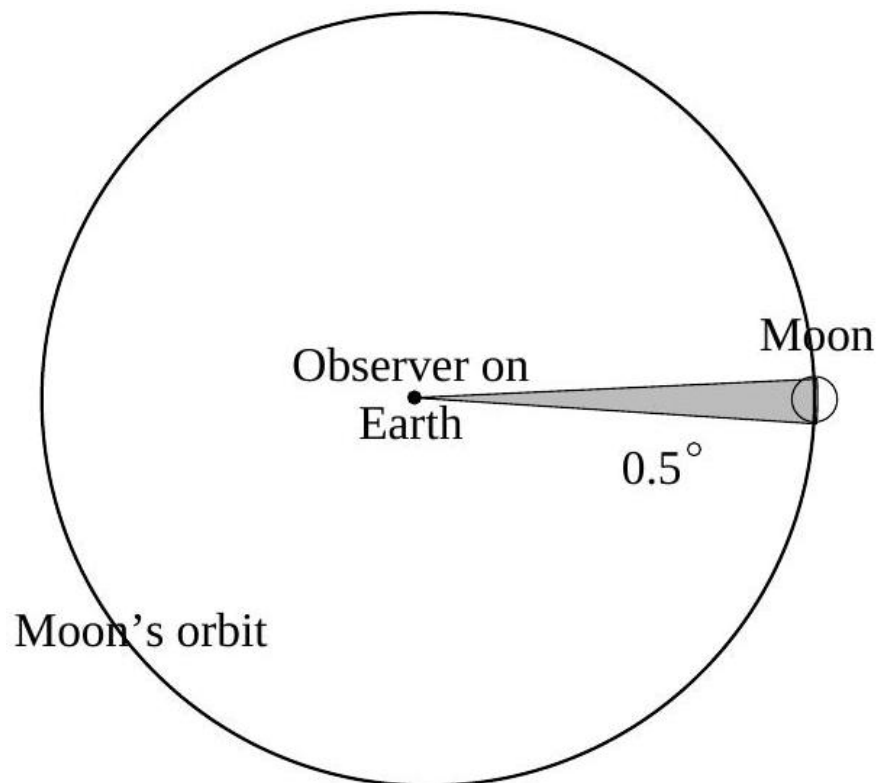


Figure 5: The orbit of the Moon as seen from above.

Use the two photographs of the Moon taken during the Lunar Eclipse of October 27, 2004 (see Figure 6) to measure the diameter of the Moon and the length of the Moon's orbit. Use the perpendicular bisector method to find the center of Earth's shadow in the image s and measure its diameter. Then measure the diameter of the Moon in the image m using the same method. Do this for both images.

For Image 1, record your values here:

$$\begin{aligned}s_1 &= _____ \\ m_1 &= _____ \\ \frac{s_1}{m_1} &= _____\end{aligned}$$

For Image 2, record your values here:

$$\begin{aligned}s_2 &= _____ \\ m_2 &= _____ \\ \frac{s_2}{m_2} &= _____\end{aligned}$$

Now calculate the average ratio of s and m from your measurements in Figure 6 and record the value here.

The average ratio, $\frac{s}{m} = _____$. Note that the ratio in image is the same as the ratio in the real world, i.e. $\frac{S}{M} = \frac{s}{m}$, so we can use this average ratio to calculate the diameter of the Moon and the distance to the Moon.

Now use Fig. 5 to go from the diameter of the Moon to the circumference of the Moon's orbit. The diameter of the Moon is seen under 0.5° angle from Earth, and the complete orbit of the Moon is seen under 360° angle from Earth. Use this information to calculate the circumference of the Moon's orbit; record your value in Table I.

	Angle ($^\circ$)	Length (km)
M	0.5°	
Circumference of the orbit of the Moon	360°	

Table I: Circumference of the orbit of the Moon.

After obtaining the circumference, now calculate the radius of the Moon's orbit, R . Use the fact that the circumference of a circle is equal to $2\pi R$, where $\pi = 3.14$. Finally, list your values in Table II and compare them with the accepted values. Are your values and the accepted ones much different, calculate and record the percent deviations;

$$\text{Percent Deviation} = \left| \frac{\text{Your Value} - \text{Accepted Value}}{\text{Accepted Value}} \right| \times 100\%.$$

	Accepted Value (km)	Your Value (km)	Percent Deviation (%)
Diameter of the Moon M	3,476		
Earth-Moon distance R	384,400		

Table II: Diameter of the Moon and the Earth-Moon distance; final results.

Image 1

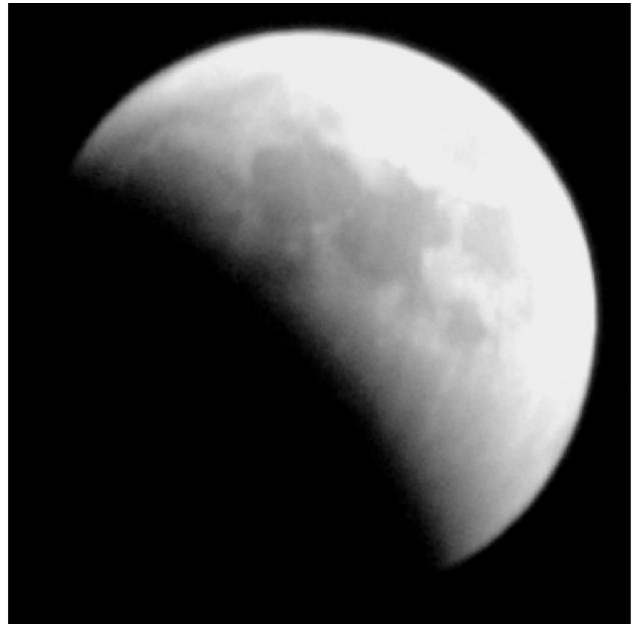


Image 2



Aim

Results

Conclusion
